Grade 4 Science

Unit 2 Assessment — Using Natural Resources for Energy

Name: _____ HR: 4A 4B 4C 4D

Directions: Read each question and choose the ONE BEST ANSWER.

Section 1: Vocabulary Matching (Questions 1–8)

Write the letter of the definition that matches each word. Note: Two definitions will not be used.

Vocabulary	Definitions
1. _____ **Renewable Resource**	**A.** A natural resource that cannot be restored as quickly as it is used by humans.
2. _____ **Nonrenewable Resource**	**B.** A machine that helps produce electricity when a turbine spins.
3. _____ **Fossil Fuel**	**C.** A liquid fossil fuel formed deep underground that is refined into gasoline and diesel.
4. _____ **Petroleum**	**D.** Heat transferred from Earth's interior.
5. _____ **Uranium**	**E.** An energy-dense fuel formed from the fossilized remains of ancient living organisms.
6. _____ **Turbine**	**F.** A natural resource that can be restored or replenished as fast as or faster than it is used.
7. _____ **Generator**	**G.** The energy of moving water flowing through rivers or ocean tides.
8. _____ **Solar Energy**	**H.** Energy from sunlight that can be used to produce electricity.
	I. A machine with blades that spins when moving air, water, or steam pushes against it.
	J. A mined resource used in nuclear power plants to produce heat.

Section 2: Multiple Choice (Questions 9–25)

Choose the ONE BEST ANSWER for each question.

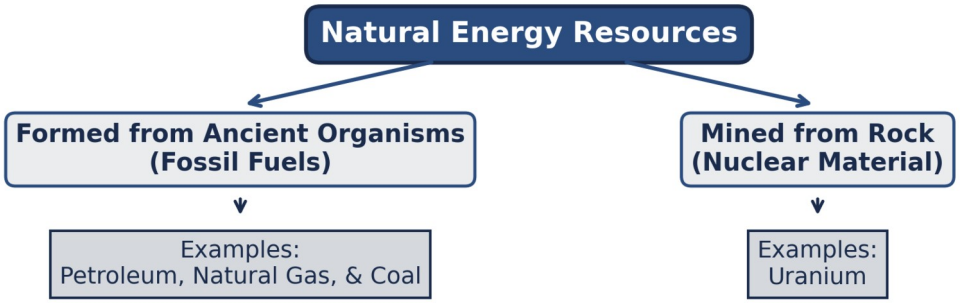
9. What is the main difference between a renewable and a nonrenewable resource?

A. Nonrenewable resources are always safer for the environment.

B. Renewable resources cause more air pollution than nonrenewable resources.

C. Renewable resources can be restored quickly, while nonrenewable resources take millions of years to form.

D. Nonrenewable resources can be restored quickly by planting trees.



10. Based on the classification chart below, which energy source should be classified as a fossil fuel?

A. Sunlight

B. Coal

C. Wind

D. Uranium

11. Why are fossil fuels like petroleum considered nonrenewable resources?

A. They are made in factories every day.

B. They are being used much faster than nature can create them.

C. They can never be completely used up.

D. They are found everywhere on Earth's surface.

12. Which fossil fuel is a liquid formed underground that is refined into gasoline and diesel?

A. Petroleum (Crude Oil)

B. Coal

C. Natural Gas

D. Uranium

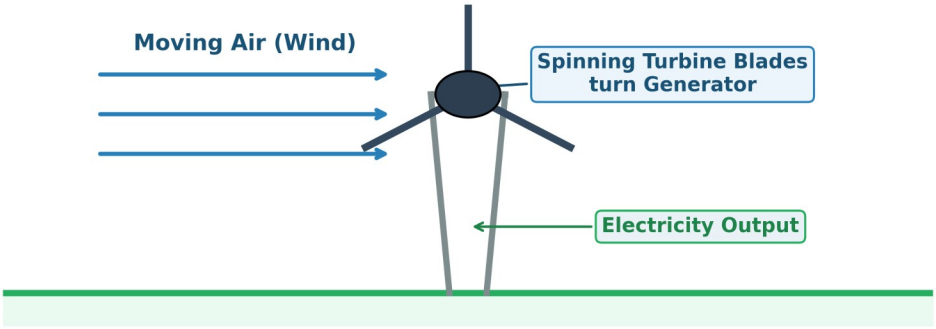
13. Nuclear energy plants use uranium as a resource. What happens next in the energy pathway?

A. Uranium produces heat that helps create steam

B. Uranium turns directly into electricity

C. Uranium absorbs sunlight

D. Uranium moves water through rivers



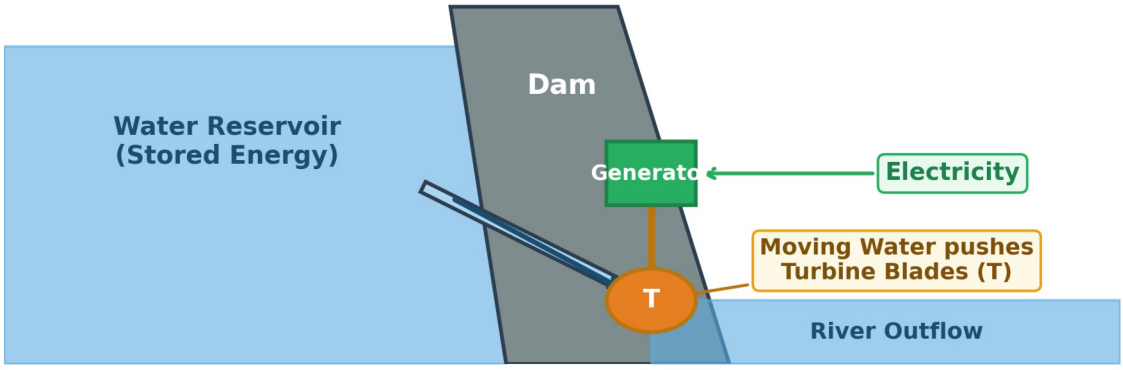
14. The wind turbine in the diagram converts the energy of motion of moving air into what final form of energy?

A. Geothermal energy

B. Electricity

C. Heat energy

D. Solar energy



15. How does a hydroelectric dam generate electricity?

A. The energy of moving water pushes a turbine connected to a generator.

B. The dam burns water to create steam.

C. The dam uses solar panels to heat the water.

D. The water creates wind that blows across the dam.

16. A town has strong sunlight most days but very little moving water. Which energy resource would be most reliable based on availability?

A. Hydroelectric power

B. Coal power

C. Wind power

D. Solar power

17. What is one major limitation (cost) of using solar energy?

A. It produces large amounts of smoke and air pollution.

B. It requires millions of years to form underground.

C. Solar energy depends on available sunlight, so less sunlight can limit electricity production.

D. It requires digging a deep underground coal mine.

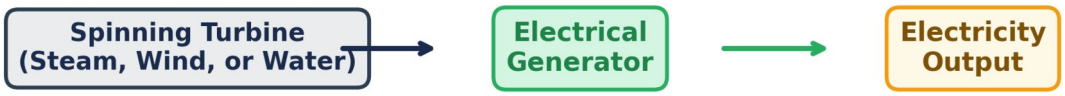
18. Geothermal energy relies on heat that comes from where?

A. The sun's surface rays

B. Warm ocean surface waves

C. Burning fossilized plants

D. Deep inside Earth



19. In most power plant systems, a spinning turbine turns a generator. What is the final form of energy produced by the generator?

A. Electricity

B. Sound energy

C. Heat energy

D. Solar energy

20. A hybrid car is engineered to use two different sources of energy. Which combination is commonly used in hybrid vehicles?

A. Coal combustion and natural gas

B. Gasoline fuel and battery power

C. Wind power and solar energy

D. Uranium nuclear fuel and water

21. Propane gas is abundant in tanks today, but it takes millions of years to form deep underground alongside crude oil. This evidence shows that propane gas is a:

A. Renewable resource

B. Biofuel

C. Nonrenewable resource

D. Geothermal resource

22. Which of the following is a major environmental cost associated with burning fossil fuels like coal and oil?

A. Releasing smoke and air pollutants into the air

B. Cooling down Earth's ocean temperatures too fast

C. Creating excessive wind currents around power plants

D. Depleting the total amount of sunlight reaching Earth

23. Scientists are researching methane hydrate ('fire ice') frozen deep under ocean floors. Because it is a fossil fuel, scientists know that methane hydrate is:

A. Easy to replace quickly after it is burned

B. A nonrenewable energy resource

C. Powered directly by solar panels

D. Completely free of any environmental costs



24. Although coal power plants and nuclear power plants use different fuels, what shared step do both use to generate electricity?

A. They both burn fossil fuel plants underground.

B. They both use rooftop solar cells to heat water.

C. They both use heat to turn water into steam that spins a turbine generator.

D. They both rely directly on flowing river currents.

25. A mountain town wants to switch to renewable energy. The area has strong, steady wind gusts every day, but it is cloudy most of the year and has no large rivers. Which renewable resource is the most reliable choice for this town?

A. Hydroelectric power

B. Solar power

C. Coal energy

D. Wind power